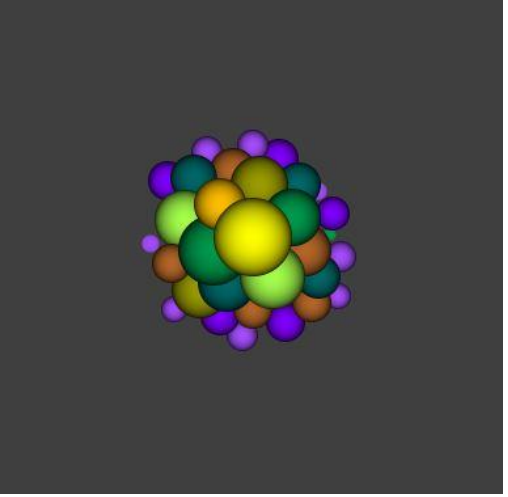
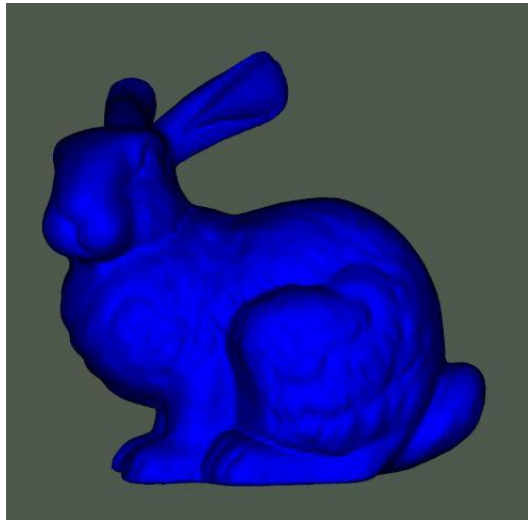


CS 440: Computer Graphics

Week 1

Lecture 2: Math

Acknowledgements: Cem Yuksel, University of Utah; Daniel Ritchie, Brown University; Waqar Saleem, Habib University



Announcements and Recap

- Syllabus Updates
- Canvas Updates
- Overview of Computer Graphics
- Applications

Quick Survey

- Let’s see what everyone’s thoughts are before starting the course!
- Scan the QR Code:

Today: Math

- Points
- Vectors
- Matrices

Points and Vectors

- Represented as (a, b)
- Points do not have any operations with each other
- **Points are different from Vectors**
- Points can be "displaced" to create "Vectors"
  - How would we displace (0,1) to (3,3)?
- Point + Vector → Point
  - Displacing the point P by amount v, Q = P + v
  - Vector is difference of points , v = Q – P

Vectors

- Vectors are represented usually in column notation
  - $\vec{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$
- Vectors can be 1D, 2D, 3D, …nD
- Vectors have direction and magnitude (or length)
- Length

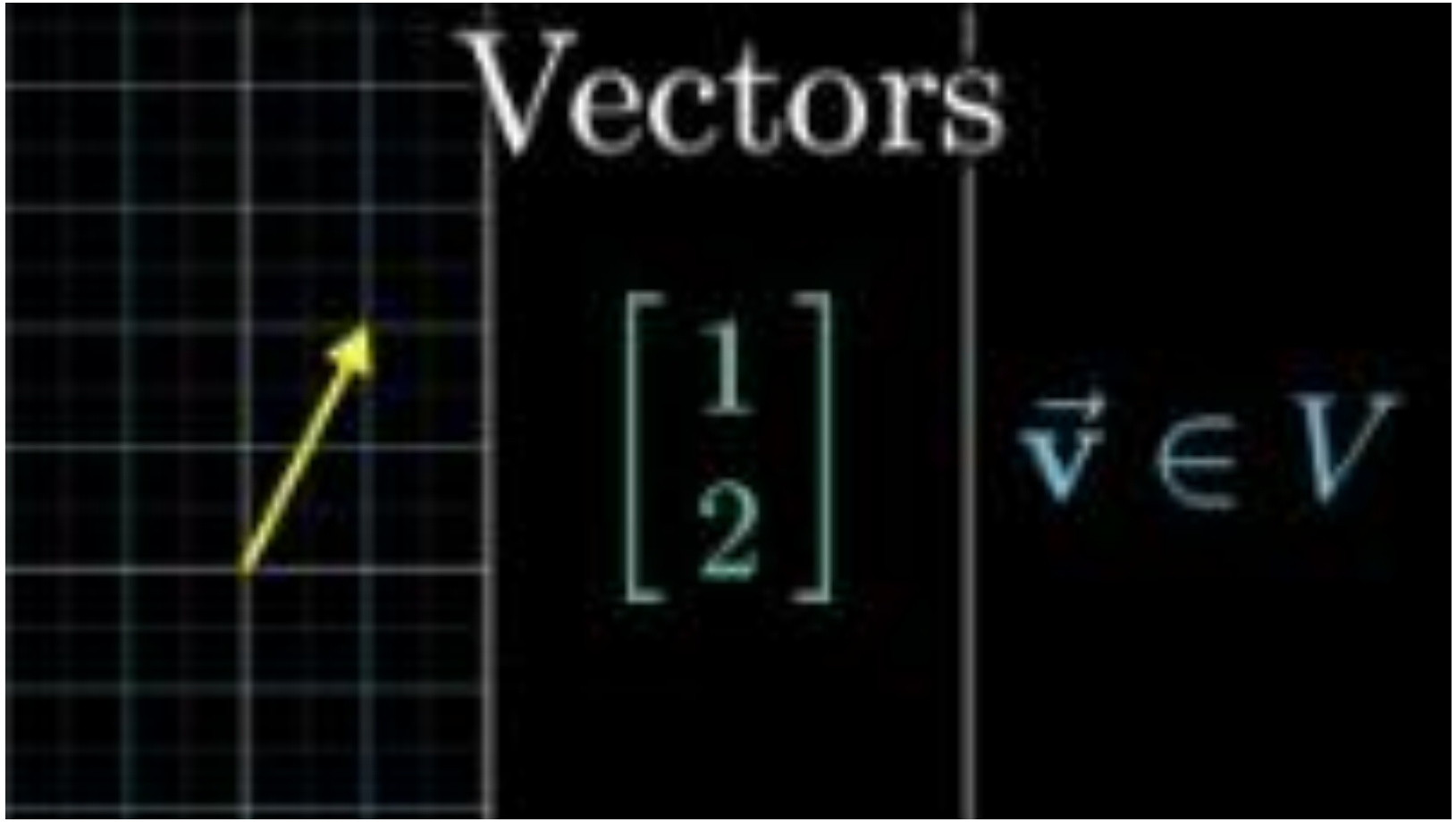
$$|v| = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

- Unit Vector

$$|v| = 1$$

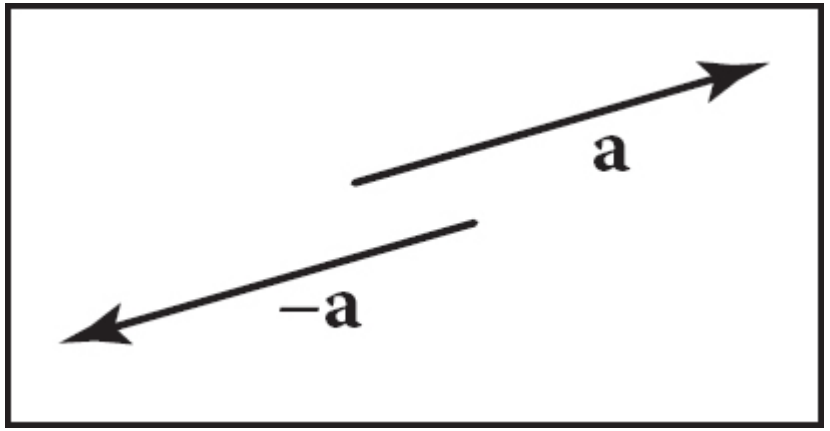
Vectors

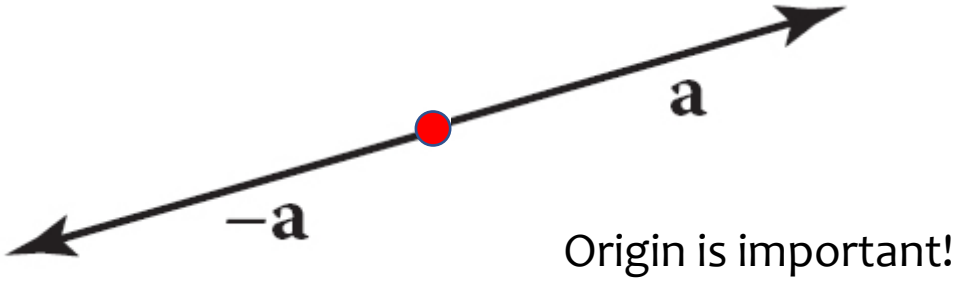
Link: https://youtu.be/fNk_zzaMoSs?si=fqKmq90x0YQ_k51r&t=120



Vector Operations

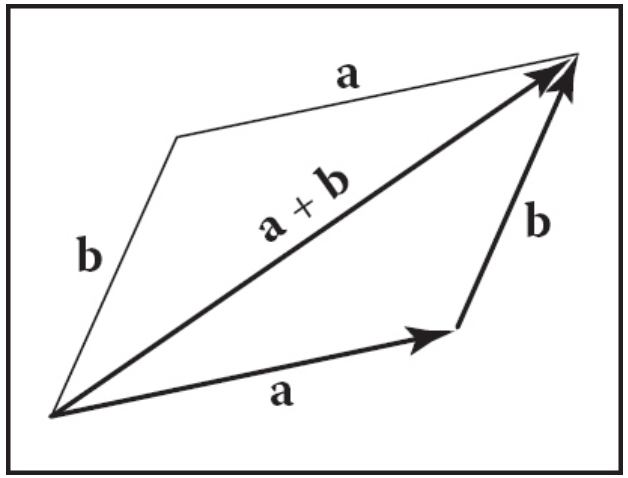
- **Inverse / Negative Vector**

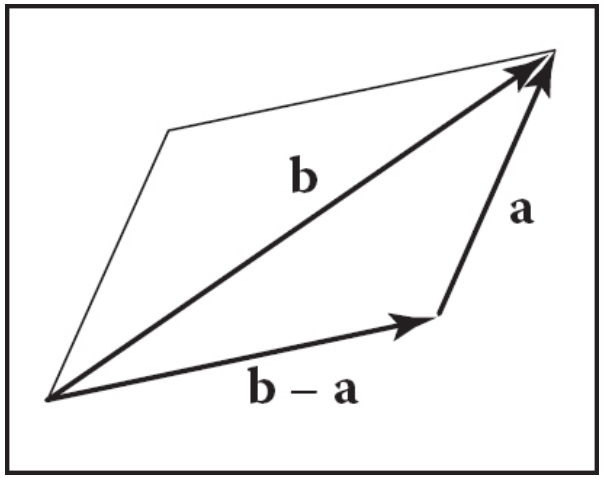




Vector Operations

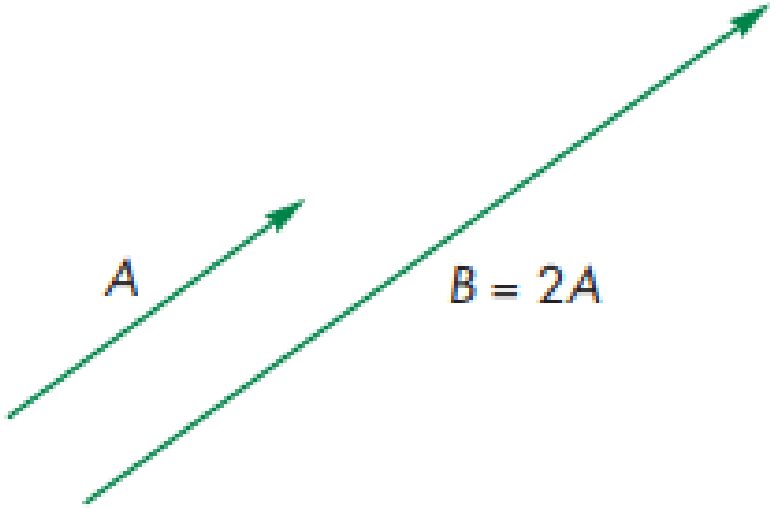
- **Vector Addition / Subtraction**
  - Parallelogram rule
  - Sum of two vectors is found by placing the tail of either vector against the head of the other





Vector Operations

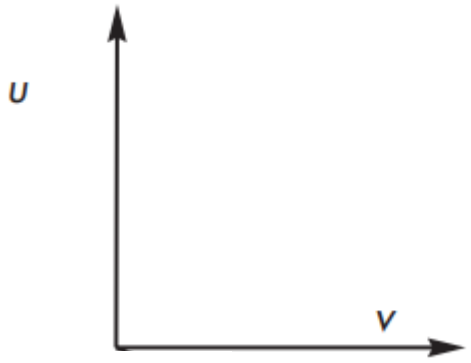
- **Scalar Multiplication**
  - Scalar – a single numerical value that belongs to a certain set
  - E.g. Real or complex numbers

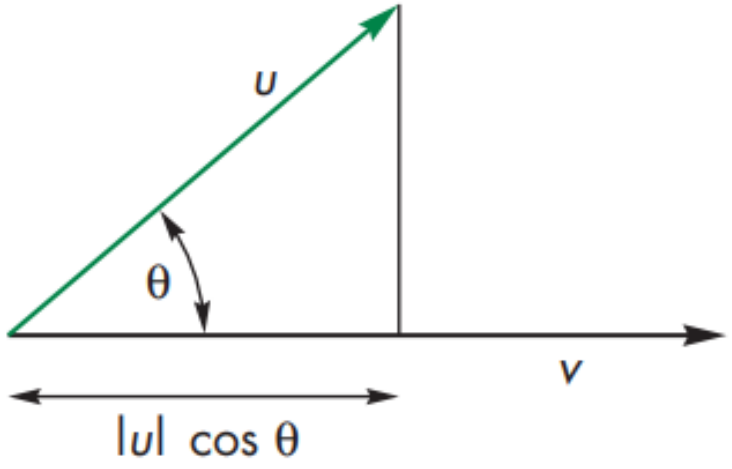


Vector Operations

- **Dot product (or inner product)**
  - Takes two vector, products a **scalar**
  - If $u \cdot v = 0$, then the vectors are **orthogonal**
  - Angle between two vectors: $\cos\theta = \frac{u \cdot v}{|u||v|}$
  - Orthogonal projection: $|u|\cos\theta = \frac{u \cdot v}{|v|}$
  - In coordinate form:

    $$u \cdot v = u_x v_x + u_y v_y + u_z v_z$$

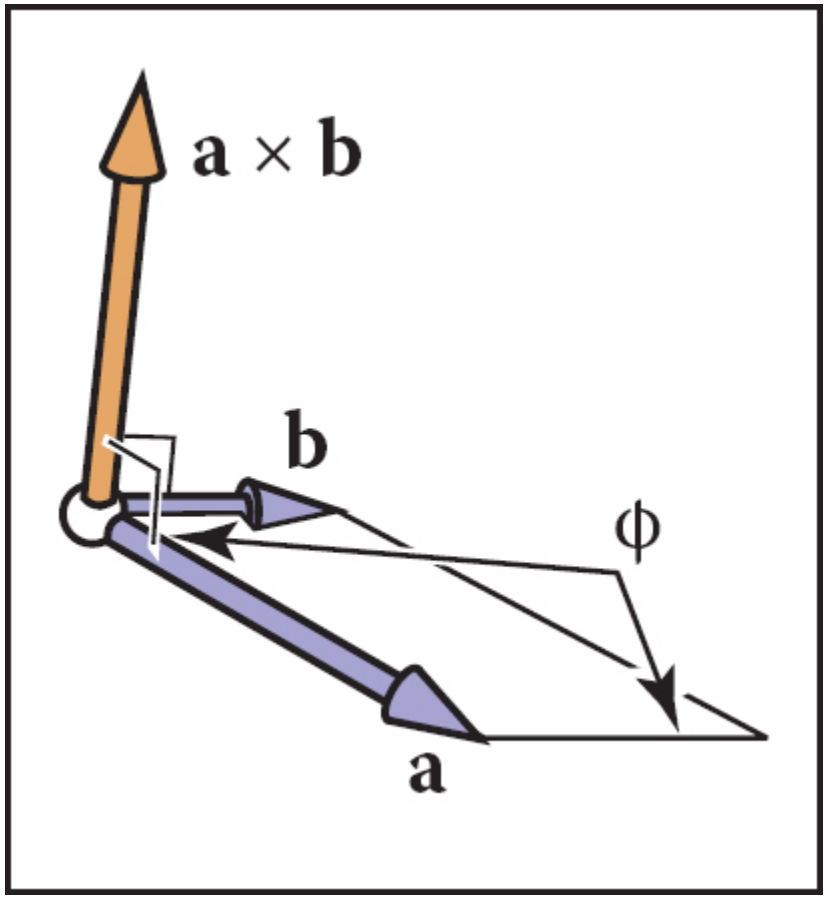




Vector Operations

- **Cross product**
  - **a x b** is a vector that is perpendicular to both **a** and **b**
  - Length of **a x b** is the area of between the two vectors
  - Angle: $\sin\theta = \frac{|a \times b|}{|a||b|}$
  - In coordinate form:

$$a \times b = (a_y b_z - a_z b_y, a_z b_x - a_x b_z, a_x b_y - a_y b_x)$$



Matrices

- Matrices can be 2D, 3D, …, nD
- Vector can be considered as a special case of matrix
- A 2x2 matrix $\begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix}$ can be viewed as
  - Two column vectors, $\begin{bmatrix} 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 3 \\ 9 \end{bmatrix}$
  - Two row vectors, $\begin{bmatrix} 2 & 3 \end{bmatrix}, \begin{bmatrix} 6 & 9 \end{bmatrix}$

Matrix Operations

- Transpose
  - $\begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix}^T = \begin{bmatrix} 2 & 6 \\ 3 & 9 \end{bmatrix}$

- Matrix Multiplication

$$Ab = \begin{bmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = \begin{bmatrix} a_{00}b_x + a_{01}b_y + a_{02}b_z \\ a_{10}b_x + a_{11}b_y + a_{12}b_z \\ a_{20}b_x + a_{21}b_y + a_{22}b_z \end{bmatrix}$$

- Not commutative, $AB \neq BA$